



# Chapter 1: Introduction to Derivatives Markets

## 1. The Growing Importance and Definition of Derivatives

### a. Derivatives Market Overview and Importance

- **Rapid Growth:** Derivatives have become increasingly vital in finance over the last 40 years.
- **Active Trading:**
  - **Exchanges:** Futures and options are actively traded globally on exchanges.
  - **Over-The-Counter (OTC) Market:** Forward contracts, swaps, options, and other customized derivatives are entered into by financial institutions, fund managers, and corporations.
- **Widespread Application:** Derivatives are used in various contexts:
  - Added to **bond issues**.
  - Used in **executive compensation** plans.
  - Embedded in **capital investment opportunities** (real options).
  - Used to **transfer risks** in mortgages from lenders to investors (securitization).
- **Market Size:** The derivatives market is **massive**—significantly larger than the stock market, when measured by underlying assets.
  - The value of assets underlying outstanding derivatives is **several times the world's gross domestic product (GDP)**.
- **Key Role:** Derivatives play a crucial role in transferring a wide range of risks across the economy from one entity to another.

### b. Definition of a Derivative

- **Definition:** A financial instrument whose **value depends on (or derives from)** the values of other, more basic, **underlying variables**.
- **Underlying Variables:**
  - Most often, the prices of **traded assets** (e.g., a stock option's value depends on the stock price).
  - Can be dependent on almost **any variable**, from commodities (hogs) to meteorological data (amount of snow).

## c. Market Developments Since 1988

- **New Products:** Active trading has emerged in:
  - Credit derivatives
  - Electricity derivatives
  - Weather derivatives
  - Insurance derivatives
  - New types of interest rate, foreign exchange, and equity derivative products.
- **Risk Management:** New ideas in risk management and risk measurement.
- **Real Options:** Capital investment appraisal often involves the evaluation of "real options."
- **New Regulations:** Introduction of many new regulations for OTC derivatives markets.

## d. Role in the 2007 Credit Crisis and Regulation

- **The Crisis:** Derivatives markets faced heavy criticism for their role in the credit crisis that began in 2007.
  - **Securitization:** Derivative products were created from portfolios of risky US mortgages using this process (explained in detail in Chapter 8).
  - The value of these products collapsed when house prices declined, causing massive losses for financial institutions and investors.
- **Increased Regulation:** As a result of the crisis, derivatives markets are now **more heavily regulated**.
  - Banks must keep **more capital** for risks.
  - More attention is paid to **liquidity**.
- **Valuation Evolution:** Banks have changed how they value derivatives:
  - More attention is given to **collateral arrangements** and **credit issues**.
  - Many banks have changed the proxies for the "risk-free" interest rate to reflect their **funding costs** (discussed in Chapter 9 and Chapter 24).
- **Usage Summary:** Derivatives can be used for **hedging, speculation, or arbitrage** (further elaborated later).

## 2. Exchange-Traded Markets (ETM)

### a. History and Key Exchanges

- **Definition:** A market where individuals trade **standardized contracts** defined by the exchange.
- **Chicago Board of Trade (CBOT) - 1848:** Established to standardize grain quantities/qualities. Led to the first futures-type contract (the **to-arrive contract**).
- **Chicago Mercantile Exchange (CME) - 1919:** A rival futures exchange.

- **CME Group:** CBOT and CME have merged, now including NYMEX, COMEX, and KCBT.
- **Chicago Board Options Exchange (CBOE) - 1973:** Started trading standardized **call option** contracts on 16 stocks. **Put option** contracts started in 1977.

## b. The Role of the Clearing House

- **Function:** Stands between the two traders to manage risks.
- **Example:** If Trader A agrees to buy gold from Trader B, A contracts to buy from the clearing house, and B contracts to sell to the clearing house.
- **Credit Risk Mitigation:** Traders do not worry about the counterparty's creditworthiness.
- **Margin:** The clearing house requires both traders to deposit funds (known as **margin**) to ensure they meet their obligations (discussed in Chapter 2).

## c. Electronic Markets

- **Shift from Open Outcry:** Traditional "open outcry" (shouting and hand signals on the floor) has been largely replaced by **electronic trading**.
- **Electronic Trading:** Traders enter desired trades at a keyboard, and a computer matches buyers and sellers.
- **Algorithmic Trading:** Electronic trading has fueled the growth of **high-frequency and algorithmic trading**, where computer programs initiate trades, often without human intervention.

# 3. Over-The-Counter Markets (OTC)

## a. Participants and Trading

- **Participants:** Banks, other large financial institutions, fund managers, and corporations.
- **Bilateral Trading:** Traditionally, trades involved direct contact (phone, email) or use of an **interdealer broker**.
- **Market Makers:** Banks often act as market makers for commonly traded instruments, quoting both a **bid price** (to buy) and an **offer price** (to sell).

## b. Clearing and Agreements

- **Central Counterparty (CCP):** Parties can present the trade to a CCP (similar to an exchange clearing house), which stands between them to mitigate credit risk.
- **Bilateral Clearing:** Parties settle trades directly. They typically sign an agreement covering:
  - Circumstances for terminating outstanding transactions.
  - Calculation of settlement amounts upon termination.

- Calculation of **collateral** (if any) to be posted by each side.
- (Discussed further in Chapter 2 and Chapter 24).

## c. Post-Credit Crisis Regulation (Forced Exchange-Like Structure)

- **Prior to 2007:** OTC markets were largely unregulated.
- **New Regulations:** Regulations were introduced to improve **transparency, efficiency, and reduce systemic risk** following the credit crisis and the Lehman Brothers bankruptcy (Business Snapshot 1.1 & 1.2).

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- **Three Key Changes:**
  - i. **Swap Execution Facilities (SEFs):** Standardized OTC derivatives in the US must, whenever possible, be traded on SEFs—platforms for posting/accepting bid and offer quotes.
  - ii. **CCP Requirement:** Most parts of the world now require a CCP for most standardized derivatives transactions.
  - iii. **Central Registry Reporting:** All trades must be reported to a central registry.

**Business Snapshot 1.2 Systemic Risk:** The risk that a default by one financial institution will trigger a ripple effect, leading to defaults by others and threatening the stability of the entire financial system. Regulators are deeply concerned about this, leading to government bailouts in the 2007–2008 crisis.

## d. Market Size Comparison

- **OTC vs. Exchange-Traded:** The OTC market is **much larger** than the exchange-traded market.
- **Volume vs. Size:** OTC has fewer transactions per year, but the **average size** of transactions is much greater.
- **BIS Statistics (Dec 2012):**
  - **OTC Total Principal:** \$632.6 trillion
  - **Exchange-Traded Underlying Assets:** \$52.6 trillion
- **Important Distinction (Principal vs. Value):** The total **principal amount** underlying an OTC transaction (e.g., \$100 million) is not the same as its **market value** (which might be only \$1 million).
  - The Bank for International Settlements (BIS) estimated the **gross market value** of all OTC transactions in December 2012 to be about **\$24.7 trillion**.